

like SOC, SOX, HIPAA, etc, depending on local compliance requirements

- Cloud migration, backup and disaster recovery to enable faster cloud adoption and transformation to the target cloud environment
- Monitoring and observability to enable support functionalities and cloud management services, including native and third party monitoring integration
- Cloud management and resource optimisation to enable better total cost of ownership (TCO) and effective return on investment in cloud adoption.

Reference architecture of multi-cloud management

When enterprises begin migrating to the cloud, they tend to move slowly so that they can recover quickly in case the migration strategy fails. Hybrid cloud management plays a vital role here. A hybrid cloud has architecture designs of two different types – legacy applications remain on-premises due to data sensitivity or other reasons, and modern applications are moved to the cloud.

Keeping legacy applications on-premises and making them communicate with the applications in the cloud using APIs or asynchronous message can be one design approach. In another approach, on-premises applications, as well as the cloud infrastructure and platform are set up as close as possible in order to have seamless integration between them.

Appliance services in AWS, MS Azure and GCP (Google Cloud Platform), like AWS Outposts, Azure Stack or Google Anthos, can help with the second approach. They help in tight integration of services in hybrid cloud environments, and also provide a centralised infrastructure management and monitoring facility. They are helpful for easy lift-and-shift of on-premises applications to a public cloud at a later point in time.

Google Anthos is powerful as it supports a hybrid cloud as well as a multi-cloud environment with an agnostic design. Azure enables seamless integration of services between on-premises and public cloud applications, while AWS outposts help by quickly provisioning an infrastructure setup by the cloud service provider itself.

Agnostic cloud management solutions

Though there are many multi-cloud management tools available in the market like Apptio Cloudability, CloudHealth, Matilda, Densify, and CloudChecker, to name a few, there are many important features that are covered in native services itself. For example, GCP has cloud management services for reporting, dashboard, API integration, datalink, budget and alerts, and advisory services integrated together.

If you are looking for an agnostic cost management service for multi-cloud scenarios, you can go for third party tools like Apptio Cloudability in order to have a uniform management policy, strategy and services across different cloud environments.

On the other hand, when you are looking for lower capital expenditure (CAPEX) to avoid licensing costs for third party cost management tools and want to get quickly into cost management, then you can choose a GCP native management tool or any native management service.

Cloud service expense management (CSEM), also called FinOps, is a new-age requirement from cloud service platforms that helps in management of cloud service costs. Typically, there is transparency with respect to cloud service usage patterns. Various manual and automated facilities are now offered by cloud service providers to manage the costs associated with their services, as well as enable cost management through budgets, alerts and platform service provisioning.

Feature	 AWS Outposts	 Azure Stack	 Google Anthos
Cloud and On-premises management			
Multi-cloud support and vendor agnostic architecture			
Edge support for local caching/processing of data			
Data encryption in hybrid services (on-premises <-> cloud)			
Infrastructure setup and installation by Cloud Service Provider			
On-premises supporting extension of public cloud services			
Serverless workload/FaaS support			
IaC support			

Figure 1: Feature comparison of cloud management activities across CSPs